Session Objectives

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1. Describe the opioid analgesic drug classes; including full and partial agonists.
2. Identify the specific drugs. Correlate the use of each drug to the mechanism of action.
3. Describe the specific adverse effects associated with the use of opioids.
4. Identify how differences in pharmacokinetics can enhance adverse effects.
5. Recognize the signs and symptoms of an opioid overdose. Differentiate an opioid overdose from withdrawal.
6. Describe the critical use of opioid antagonists in overdose situations.

Source: Course Syllabus

Analgesia

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- **Strong agonists: morphine, methadone, heroin, meperidine, fentanyl**
- **Mixed agonist-antagonist: buprenorphine**
- **Weak (or partial) agonist: codeine, hydrocodone (often combined with COX inhibitor).**

Cardio-Respiratory

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- Vasodilation= hypotension
- Reduction of myocardial oxygen demand, cardiac work.
- Decrease PVR, preload, afterload
- Cause of death in overdose =Respiratory depression
- Opioids decrease the sensitivity of the medulla to increased carbon dioxide.
- Increase intracranial pressure (worsen head injury outcome).

Neonatal Abstinence Syndrome

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- The fetus may become physically dependent in utero and manifest withdrawal symptoms early postpartum
- Withdrawal symptoms in the infant: irritability, shrill crying, diarrhea, or seizures.
- More severe withdrawal: camphorated tincture of opium (previous). <u>Current: Morphine, methadone</u>
- Phenobarbital and clonidine (adjunctive use)

Opioid Withdrawal

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- Occurs 6-12 h after last dose, continues for several days, and ends within 72 h.
- Symptoms: Rhinorrhea, sweating, lacrimation, chills, gooseflesh, yawning, anxiety, aggression, mydriasis
- Drugs with a long $t_{1/2}$, produce a prolonged withdrawal (Methadone)
- Opioids bind in the nucleus accumbens, release dopamine leading to addiction "reward center"

Morphine

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- “Standard” opioid agonist
- Reacts with all opioid receptors
- Uses: moderate-severe pain, reduces dose of anesthetics, and anxiolytic effect.
- MI: produces vasodilation
- Hypersensitivity in respiratory disease
- Respiratory depression (fatal)
- Not for head trauma patients
- Intense sedation and respiratory depression when combined with other sedatives, alcohol, or barbiturates.

Meperidine

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- Rapid onset-short duration of action.
- Normeperidine (metabolite) can induce CNS stimulation.
- Serotonin syndrome with SSRIs, linezolid, tricyclic antidepressants, "triptan" drugs for migraines.
- Not with monoamine oxidase inhibitors (MAOIs)
- Anti-muscarinic effects (mydriasis, tachycardia)
- Some local anesthetic effects

Levorphanol

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- L-isomer of morphine
- Moderate to severe pain
- D-isomer (Dextromethorphan; DM). Not addictive cough suppression at normal doses.
  - NMDA receptor antagonist
  - SERT and NET reuptake inhibition

Heroin: Schedule I

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- Injected, snorted, adulterated.
- Chronic injection leads to collapse of vessels
- Highly lipophilic
- Metabolized to morphine ~diacetylmorphine
- Pregnancy: newborn addicted to heroin, withdrawal.
- Increased incidence of HIV and hepatitis (needle sharing)
- Oxycontin®- gave a "heroin like" effect when crushed and snorted. New formulations provide special coatings to prevent crushing.

Tramadol

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- μ receptor agonist and reuptake inhibitor of NE and 5HT
- Can precipitate withdrawal.
- CYP2D6 polymorphism "ultra-fast metabolism"
- Used for chronic pain (neuropathic)
- Combined with acetaminophen/NSAIDs
- Tapentadol (μ receptor agonist and reuptake inhibitor of NE, minimal effect on 5-HT).
- Both have serotonin syndrome warning.

Methylnaltrexone

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- **Treats opioid-induce constipation.**
- **Does not cross blood brain barrier (BBB). Quaternary ammonium cation**
- **<u>Alvimopan: </u>treats postoperative ileus. Antagonist of of mu receptor in GI tract. Contraindicated in patients who had opioids within 7 days; do not give alvimopan for more than 7 days. MI risk increased.**

Mixed Agonist-Antagonist

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- Pentazocine, Butorphanol, Nalbuphine
- Retains analgesia with reduced addiction potential (κ > μ receptor).
- Less physical dependence and respiratory depression.
- Psychomimetic effects (κ receptor). Pentazocine
- Precipitate withdrawal in morphine addicted patient.

Buprenorphine

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- Partial μ agonist. At higher doses is an antagonist because of K receptor blockade
- Slow dissociation from μ receptor.
- CYP 3A4 substrate
- Sublingual, transdermal, parenteral
- Adverse effects are similar to morphine and pentazocine
- Treats opioid dependence
- Precipitate withdrawal in opioid users
- Buprenorphine and naloxone (Suboxone®)

Opioid Antagonists

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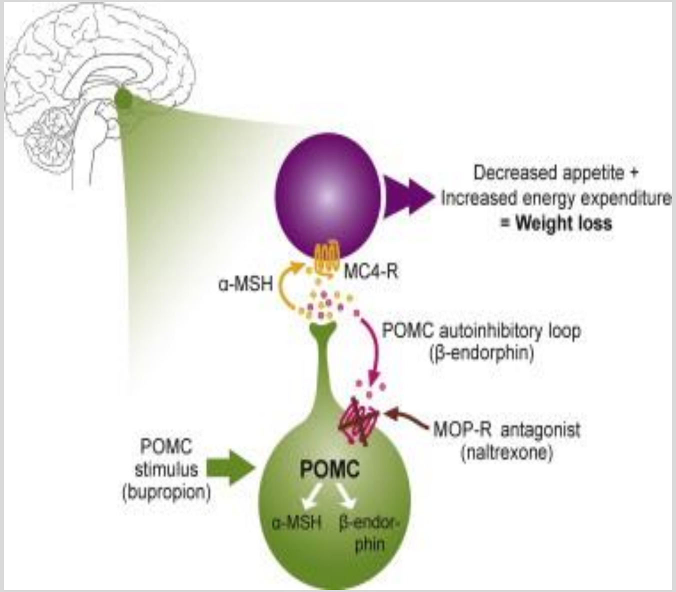
- Naloxone: IV; fast onset for opioid overdose (1-3 min). Rapid offset. Repeated administration until opioid is cleared.
- Approved in neonates with respiratory depression born to heroin mothers
- Naltrexone: oral; long-duration. Maintenance therapy. Reduces alcohol cravings. Combined with bupropion for weight loss potential.



Opioid Abuse

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- **Heroin: Rapid tolerance and dependence**
- **MOA: μ and κ receptor agonism**
- **Overdose: IV naloxone and ventilation**
- **Symptoms: "everything is down", stupor, bradycardia, decreased respirations, miosis.**
- **Withdrawal: Methadone or buprenorphine**
- **Symptoms: lacrimation, rhinorrhea, yawning, sweating, gooseflesh, tremors.**

Suzetrigine

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- Non-opioid
- MOA: Sodium channel blocker, works in periphery
- Use: moderate-to-severe acute pain, not more that 14 days.
- Adverse effects: itch, muscle spasms (increased creatine kinase), N/V.